Second virial coefficient for depletion interactions

Complex disordered systems: Colloids

Consider a dilute suspension of spherical colloidal particles of radius R_c in a solution of small ideal polymer coils of radius $R_p \ll R_c$. The polymers are non-adsorbing and behave as an ideal gas outside the colloids. The colloids interact via the **Asakura–Oosawa depletion potential** $U_{\mathrm{AO}}(r)$:

$$U_{\mathrm{AO}}(r) = \begin{cases} \infty, & r < 2R_c \\ -\Pi\, V_{\mathrm{overlap}}(r), & 2R_c \le r \le 2R_c + 2R_p \\ 0, & r > 2R_c + 2R_p \end{cases}$$

where r is the center-to-center distance, $\Pi = n_p k_B T$ is the so called *osmotic pressure* of the polymer solution, and $V_{\mathrm{overlap}}(r)$ is the overlap volume of the depletion zones, given by

$$V_{\mathrm{overlap}}\,(r) = \frac{\pi\,(4R_{\mathrm{eff}} + r)\,(2R_{\mathrm{eff}} - r)^2}{12},$$

with $R_{\mathrm{eff}} = R_c + R_p$.

1. Write the expression for the *second virial coefficient* B_2 in terms of the interaction potential:

$$B_2 = -2\pi \int_0^\infty \left(e^{-\beta U_{\mathrm{AO}}(r)} - 1\right) r^2 dr$$

2. Assuming the polymer-induced attraction is *weak* ($\beta|U_{\mathrm{AO}}(r)| \ll 1$) and that the polymers are much smaller than the colloids, expand the exponential to first order and simplify the integral.
3. Express the resulting B_2 in terms of R_c, R_p, and the polymer osmotic pressure Π.
4. Discuss qualitatively how B_2 depends on polymer concentration and size, and explain what a **negative** B_2 implies for colloidal aggregation.

i Solution

The second virial coefficient B_2 measures the effective two-body contribution to the pressure of a dilute colloidal suspension, following the virial expansion. It is defined for a pairwise potential $U(r)$ as:

$$B_2 = -2\pi \int_0^\infty \left(e^{-\beta U(r)} - 1\right) r^2 dr.$$

Here, $U(r) = U_{\mathrm{AO}}(r)$ is the Asakura–Oosawa depletion potential.

1. Rewrite B_2

   The AO potential has three regions:
   - Hard-core region ($r < 2R_c$), where $U = \infty \to e^{-\beta U} = 0$. The integrand becomes -1, giving contribution:

$$B_2^{\mathrm{HC}} = -2\pi \int_0^{2R_c} (-1) r^2 dr = 2\pi \int_0^{2R_c} r^2 dr = \frac{16\pi}{3} R_c^3,$$

which is just the hard-sphere contribution.

- Depletion region ($2R_c \le r \le 2R_c + 2R_p$), where $U(r) = -\Pi V_{\mathrm{overlap}}(r)$.
- No interaction ($r > 2R_c + 2R_p$), where $U = 0 \to e^{-\beta U} - 1 = 0$.

The overlap is written as (lecture notes)

$$V_{\mathrm{overlap}}\,(r) = \frac{4\pi}{3} R_{\mathrm{eff}}^3 \left[1 - \frac{3}{4} \frac{r}{R_{\mathrm{eff}}} + \frac{1}{16} \left(\frac{r}{R_{\mathrm{eff}}} \right)^3 \right], \quad r \le 2R_{\mathrm{eff}} = 2R_c + 2R_p$$

and can be shown to be equivalent to

$$V_{\mathrm{overlap}}\,(r) = \frac{\pi\,(4R_{\mathrm{eff}} + r)\,(2R_{\mathrm{eff}} - r)^2}{12}$$

So the integral reduces to contributions from $r < 2R_c$ (hard core) and $2R_c \le r \le 2R_c + 2R_p$ (depletion).

2. Linearize the exponential. For weak polymer-induced attraction, $\beta|U| = \beta\Pi V_{\mathrm{overlap}} \ll 1$, we can expand:

$$e^{-\beta U(r)} \approx 1 - \beta U(r) = 1 + \beta\Pi V_{\mathrm{overlap}}(r).$$

Then the integrand in the depletion region becomes:

$$-(e^{-\beta U} - 1) \approx -\beta\Pi V_{\mathrm{overlap}}(r).$$

So the contribution to B_2 from the depletion region is:

$$B_2^{\mathrm{dep}} = -2\pi \int_{2R_c}^{2R_c+2R_p} (e^{-\beta U} - 1) r^2 dr \approx -2\pi\beta\Pi \int_{2R_c}^{2R_c+2R_p} V_{\mathrm{overlap}}(r) r^2 dr.$$

3. Express B_2 in terms of R_c, R_p, Π

Combining the hard-core and depletion contributions, the second virial coefficient is

$$B_2 = \frac{16\pi}{3} R_c^3 - 2\pi\beta\Pi \int_{2R_c}^{2R_c+2R_p} V_{\mathrm{overlap}}(r)\, r^2 dr$$

We then calculate the second integral we just need to choose how to represent the overlap. We use the second expression above and leverage the fact that $R_p \ll R_c$ and that then the integral runs over $r = 2R_c + x$ with $x \in [0, 2R_p]$, so with x clearly subleading

So, then

$$4R_{\mathrm{eff}} + r = 6R_c + 4R_p + x$$

and

$$2R_{\mathrm{eff}} - r = 2R_c + 2R_p - 2R_c - x = 2R_p - x$$

So together

$$V_{\mathrm{overlap}}(x) = \frac{\pi(6R_c + 4R_p + x)(2R_p - x)^2}{12}$$

dropping subleading contributions leads to

$$V_{\mathrm{overlap}}(x) \approx \frac{6\pi R_c (2R_p - x)^2}{12}$$

Then, we rewrite

$$\int_{2R_c}^{2R_c+2R_p} V_{\text{overlap}}(r)\, r^2 dr$$

as

$$\int_0^{2R_p} V_{\text{overlap}}(x)\, (2R_c + x)^2 dx$$

and approximate as

$$\approx (2R_c)^2 \int_0^{2R_p} V_{\text{overlap}}(x) dx$$

and then

$$B_2 \approx \frac{16\pi}{3} R_c^3 - 2\pi\beta\Pi\, (2R_c)^2 \int_0^{2R_p} \frac{\pi}{2} R_c \left(2R_p - x\right)^2 dx$$

So we finally integrate

$$\int_0^{2R_p} \left(2R_p - x\right)^2 dx$$

via a change of variable $y = 2R_p - x$

$$\int_0^{2R_p} \left(2R_p - x\right)^2 dx = \int_{2R_p}^0 y^2(-dy) = \int_0^{2R_p} y^2 dy = \frac{\left(2R_p\right)^3}{3} = \frac{8R_p^3}{3}$$

which leads to

$$B_2 \approx \frac{16\pi}{3} R_c^3 - \frac{32\pi^2}{3} \beta\Pi R_c^3 R_p^3$$

where R_c is the colloid radius, R_p is the polymer radius, Π is the osmotic pressure of the polymer solution, and $V_{\text{overlap}}(r)$ is the overlap volume between two colloids at separation r. This formula shows how B_2 depends explicitly on R_c, R_p, and Π.

4. Physical interpretation

- B_2 **decreases** with increasing polymer concentration (via Π) and polymer size (R_p).
- When the depletion attraction is strong enough, B_2 can become **negative**, which signals that attractive interactions dominate over repulsion. Strictly, our result becomes negative only for $\beta\Pi R_p^3 > 1/2\pi$, where the attraction at contact, $\beta|U(2R_c)| \approx 2\pi n_p R_c R_p^2$, is already much larger than $k_B T$: this is outside the weak-attraction approximation. The exact B_2 decreases even faster, so the conclusion holds.
- Negative B_2 implies a tendency for **colloidal aggregation** or even phase separation, because the effective interaction favours close approach of colloids.